In [ ]:
```
#Phase 1: Data Loading & Initial Inspection
```

In [1]:
```
import pandas as pd
import numpy as np

# Load the dataset
df = pd.read_csv("C:\\Users\\DEBABRATA\\Downloads\\archive (10)\\financial_risk_assessment.csv")

# View basic information (data types, non-null counts)
print(df.info())

# View the first 5 rows
print(df.head())

# Check for missing values count per column
print(df.isnull().sum())

# Generate summary statistics for numerical columns
print(df.describe())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15000 entries, 0 to 14999
Data columns (total 20 columns):
 #   Column                 Non-Null Count  Dtype  
---  ------                 --------------  -----  
 0   Age                    15000 non-null  int64  
 1   Gender                 15000 non-null  object 
 2   Education Level        15000 non-null  object 
 3   Marital Status         15000 non-null  object 
 4   Income                 12750 non-null  float64
 5   Credit Score           12750 non-null  float64
 6   Loan Amount            12750 non-null  float64
 7   Loan Purpose           15000 non-null  object 
 8   Employment Status      15000 non-null  object 
 9   Years at Current Job   15000 non-null  int64  
 10  Payment History        15000 non-null  object 
 11  Debt-to-Income Ratio   15000 non-null  float64
 12  Assets Value           12750 non-null  float64
 13  Number of Dependents   12750 non-null  float64
 14  City                   15000 non-null  object 
 15  State                  15000 non-null  object 
 16  Country                15000 non-null  object 
 17  Previous Defaults      12750 non-null  float64
 18  Marital Status Change  15000 non-null  int64  
 19  Risk Rating            15000 non-null  object 
dtypes: float64(7), int64(3), object(10)
memory usage: 2.3+ MB
None
   Age      Gender Education Level Marital Status   Income  Credit Score  \
0   49        Male             PhD       Divorced  72799.0         688.0   
1   57      Female      Bachelor's        Widowed      NaN         690.0   
2   21  Non-binary        Master's         Single  55687.0         600.0   
3   59        Male      Bachelor's         Single  26508.0         622.0   
4   25  Non-binary      Bachelor's        Widowed  49427.0         766.0   

   Loan Amount Loan Purpose Employment Status  Years at Current Job  \
0      45713.0     Business        Unemployed                    19   
1      33835.0         Auto          Employed                     6   
2      36623.0         Home          Employed                     8   
3      26541.0     Personal        Unemployed                     2   
4      36528.0     Personal        Unemployed                    10   
```

```
  Payment History  Debt-to-Income Ratio  Assets Value  Number of Dependents  \
0            Poor              0.154313      120228.0                   0.0
1            Fair              0.148920       55849.0                   0.0
2            Fair              0.362398      180700.0                   3.0
3       Excellent              0.454964      157319.0                   3.0
4            Fair              0.143242      287140.0                   NaN

              City State       Country  Previous Defaults  \
0   Port Elizabeth    AS        Cyprus                2.0
1  North Catherine    OH  Turkmenistan                3.0
2      South Scott    OK    Luxembourg                3.0
3       Robinhaven    PR        Uganda                4.0
4      New Heather    IL       Namibia                3.0

   Marital Status Change Risk Rating
0                      2         Low
1                      2      Medium
2                      2      Medium
3                      2      Medium
4                      1         Low
Age                        0
Gender                     0
Education Level            0
Marital Status             0
Income                  2250
Credit Score            2250
Loan Amount             2250
Loan Purpose               0
Employment Status          0
Years at Current Job       0
Payment History            0
Debt-to-Income Ratio       0
Assets Value            2250
Number of Dependents    2250
City                       0
State                      0
Country                    0
Previous Defaults       2250
Marital Status Change      0
Risk Rating                0
dtype: int64
                Age         Income  Credit Score   Loan Amount  \
```

```
count  15000.000000   12750.000000  12750.000000  12750.000000
mean      43.452667   69933.398510    699.109098  27450.010902
std       14.910732   29163.626207     57.229465  12949.940135
min       18.000000   20005.000000    600.000000   5000.000000
25%       31.000000   44281.500000    650.000000  16352.500000
50%       43.000000   69773.000000    699.000000  27544.000000
75%       56.000000   95922.750000    748.000000  38547.500000
max       69.000000  119997.000000    799.000000  49998.000000

       Years at Current Job  Debt-to-Income Ratio   Assets Value  \
count          15000.000000          15000.000000   12750.000000
mean               9.476267              0.350438  159741.497176
std                5.769707              0.143919   80298.115832
min                0.000000              0.100004   20055.000000
25%                4.000000              0.227386   90635.250000
50%                9.000000              0.350754  159362.000000
75%               15.000000              0.476095  228707.000000
max               19.000000              0.599970  299999.000000

       Number of Dependents  Previous Defaults  Marital Status Change
count           12750.00000       12750.000000           15000.000000
mean                2.02651           1.992471               0.998467
std                 1.41130           1.416909               0.813782
min                 0.00000           0.000000               0.000000
25%                 1.00000           1.000000               0.000000
50%                 2.00000           2.000000               1.000000
75%                 3.00000           3.000000               2.000000
max                 4.00000           4.000000               2.000000
```

In [ ]:
```
#Phase 2: Data Cleaning & Preprocessing
```

In [7]:
```
# 1. Handling Missing Values
# Fill numerical missing values with median or mean
numerical_cols = ['Income', 'Credit Score', 'Loan Amount', 'Assets Value', 'Number of Dependents', 'Previous Defaults
for col in numerical_cols:
    df[col].fillna(df[col].median(), inplace=True)

# Verify no missing values remain
print(df.isnull().sum())

# 2. Encoding Categorical Variables (for Machine Learning or deeper analysis)
categorical_cols = ['Gender', 'Education Level', 'Marital Status', 'Loan Purpose', 'Employment Status', 'Payment Hist
```

```
# One-Hot Encoding or Label Encoding
# Using pd.get_dummies for one-hot encoding categorical features
df_encoded = pd.get_dummies(df, columns=['Gender', 'Education Level', 'Marital Status', 'Loan Purpose', 'Employment S
```

```
Age                     0
Gender                  0
Education Level         0
Marital Status          0
Income                  0
Credit Score            0
Loan Amount             0
Loan Purpose            0
Employment Status       0
Years at Current Job    0
Payment History         0
Debt-to-Income Ratio    0
Assets Value            0
Number of Dependents    0
City                    0
State                   0
Country                 0
Previous Defaults       0
Marital Status Change   0
Risk Rating             0
dtype: int64
```

```
C:\Users\DEBABRATA\AppData\Local\Temp\ipykernel_16068\2365502813.py:5: FutureWarning: A value is trying to be set on
a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which
we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or d
f[col] = df[col].method(value) instead, to perform the operation inplace on the original object.


  df[col].fillna(df[col].median(), inplace=True)
```

In [8]:
```
#Phase 3: Exploratory Data Analysis (EDA) & Aggregation
```

In [9]:
```
# 1. Average Income and Loan Amount by Risk Rating
risk_summary = df.groupby('Risk Rating')[['Income', 'Loan Amount', 'Credit Score', 'Debt-to-Income Ratio']].mean()
print(risk_summary)
```

```
# 2. Risk Rating distribution across Employment Status
employment_risk = pd.crosstab(df['Employment Status'], df['Risk Rating'], normalize='index') * 100
print("Employment Status vs Risk Rating (%):\n", employment_risk)

# 3. Correlation Matrix for Numerical Features
corr_matrix = df[numerical_cols].corr()
print("Correlation Matrix:\n", corr_matrix)
```

```
                  Income   Loan Amount  Credit Score  Debt-to-Income Ratio
Risk Rating
High         69130.456000  27920.201333    699.738000              0.351261
Low          69644.948667  27388.017778    698.969778              0.350018
Medium       70697.746444  27464.261556    699.123556              0.351004
Employment Status vs Risk Rating (%):
 Risk Rating             High        Low     Medium
Employment Status
Employed           10.286510  58.973339  30.740151
Self-employed      10.218393  60.508916  29.272691
Unemployed          9.492274  60.525788  29.981939
Correlation Matrix:
                        Income  Credit Score  Loan Amount  Assets Value  \
Income                1.000000      0.010759    -0.006762     -0.007365
Credit Score          0.010759      1.000000    -0.011744      0.012072
Loan Amount          -0.006762     -0.011744     1.000000     -0.010222
Assets Value         -0.007365      0.012072    -0.010222      1.000000
Number of Dependents  0.005878      0.005654     0.001370     -0.002319
Previous Defaults    -0.003737     -0.007403     0.012129      0.004805

                     Number of Dependents  Previous Defaults
Income                           0.005878          -0.003737
Credit Score                     0.005654          -0.007403
Loan Amount                      0.001370           0.012129
Assets Value                    -0.002319           0.004805
Number of Dependents             1.000000          -0.007603
Previous Defaults               -0.007603           1.000000
```

In [10]:
```
#Phase 4: Data Visualization
```

In [11]:
```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Set visual style
sns.set_theme(style="whitegrid")

# 1. Distribution of Credit Scores by Risk Rating
plt.figure(figsize=(10, 6))
sns.boxplot(x='Risk Rating', y='Credit Score', data=df, palette='Set2')
plt.title('Credit Score Distribution across Risk Ratings')
plt.savefig('credit_score_by_risk.png')
plt.show()

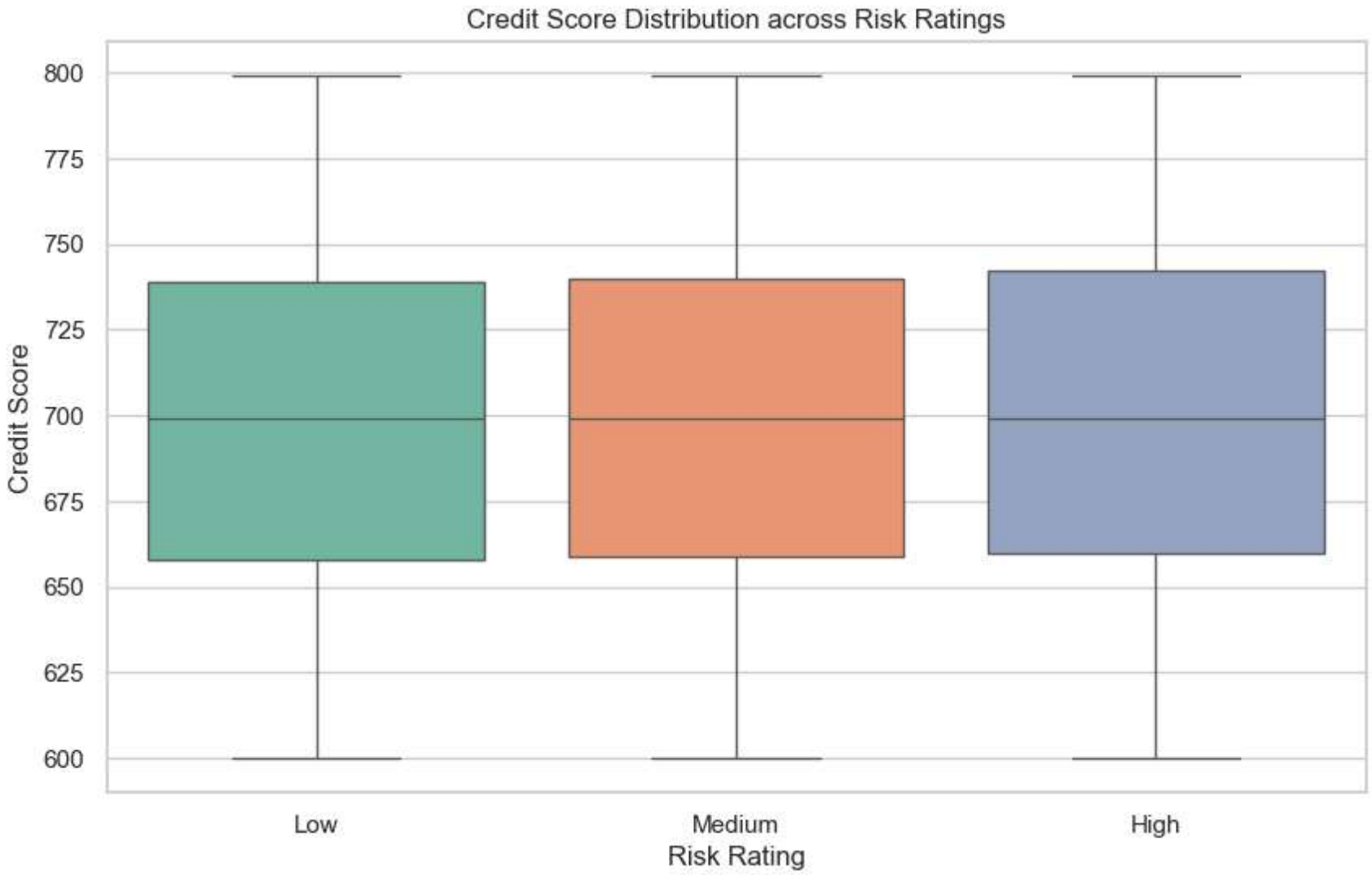
# 2. Correlation Heatmap
plt.figure(figsize=(10, 8))
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt='.2f', linewidths=0.5)
plt.title('Correlation Heatmap of Financial Features')
plt.savefig('correlation_heatmap.png')
plt.show()

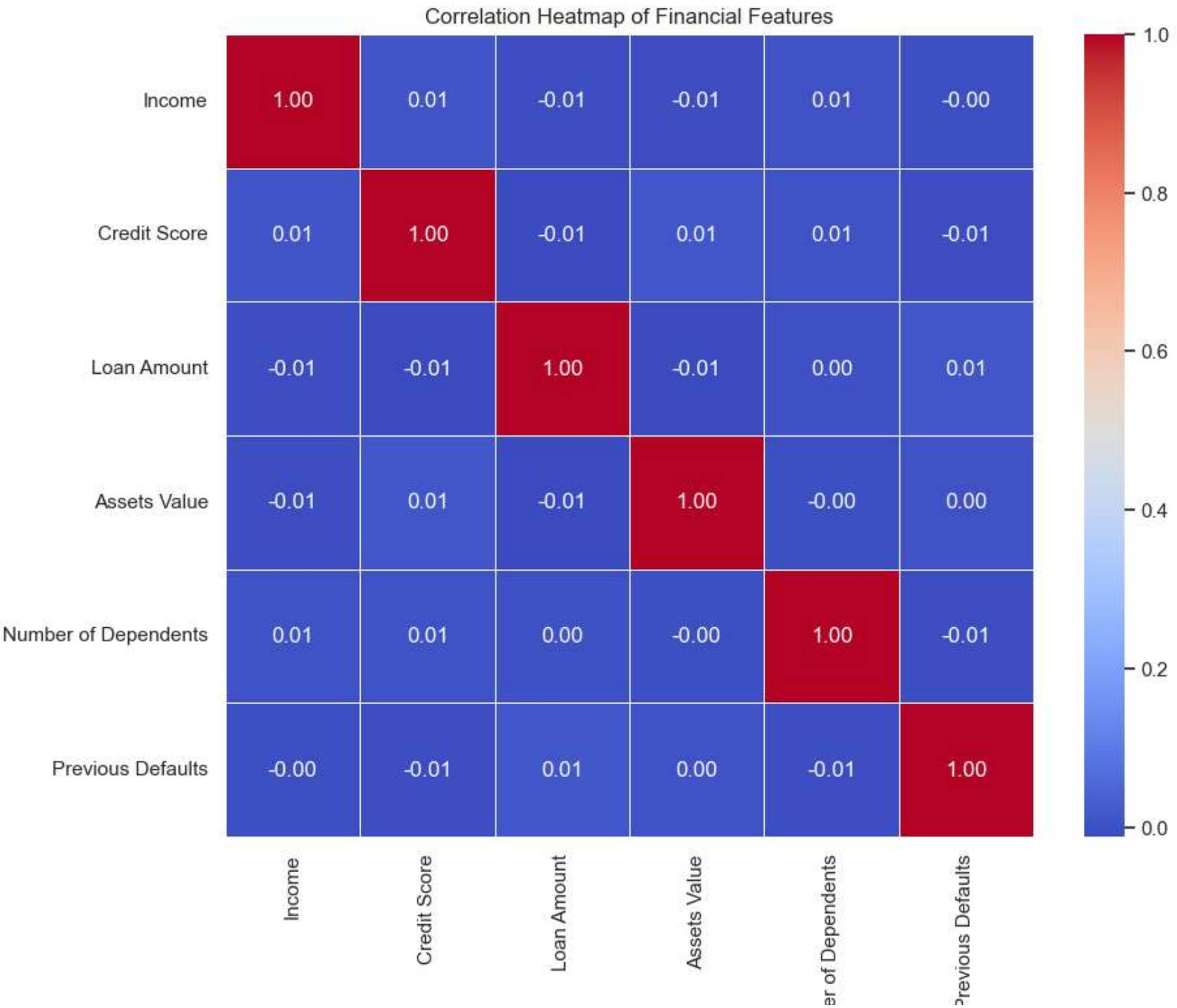
# 3. Loan Amount vs Income colored by Risk Rating
plt.figure(figsize=(10, 6))
sns.scatterplot(x='Income', y='Loan Amount', hue='Risk Rating', data=df, alpha=0.7, palette='viridis')
plt.title('Income vs Loan Amount by Risk Rating')
plt.savefig('income_vs_loan.png')
plt.show()
```

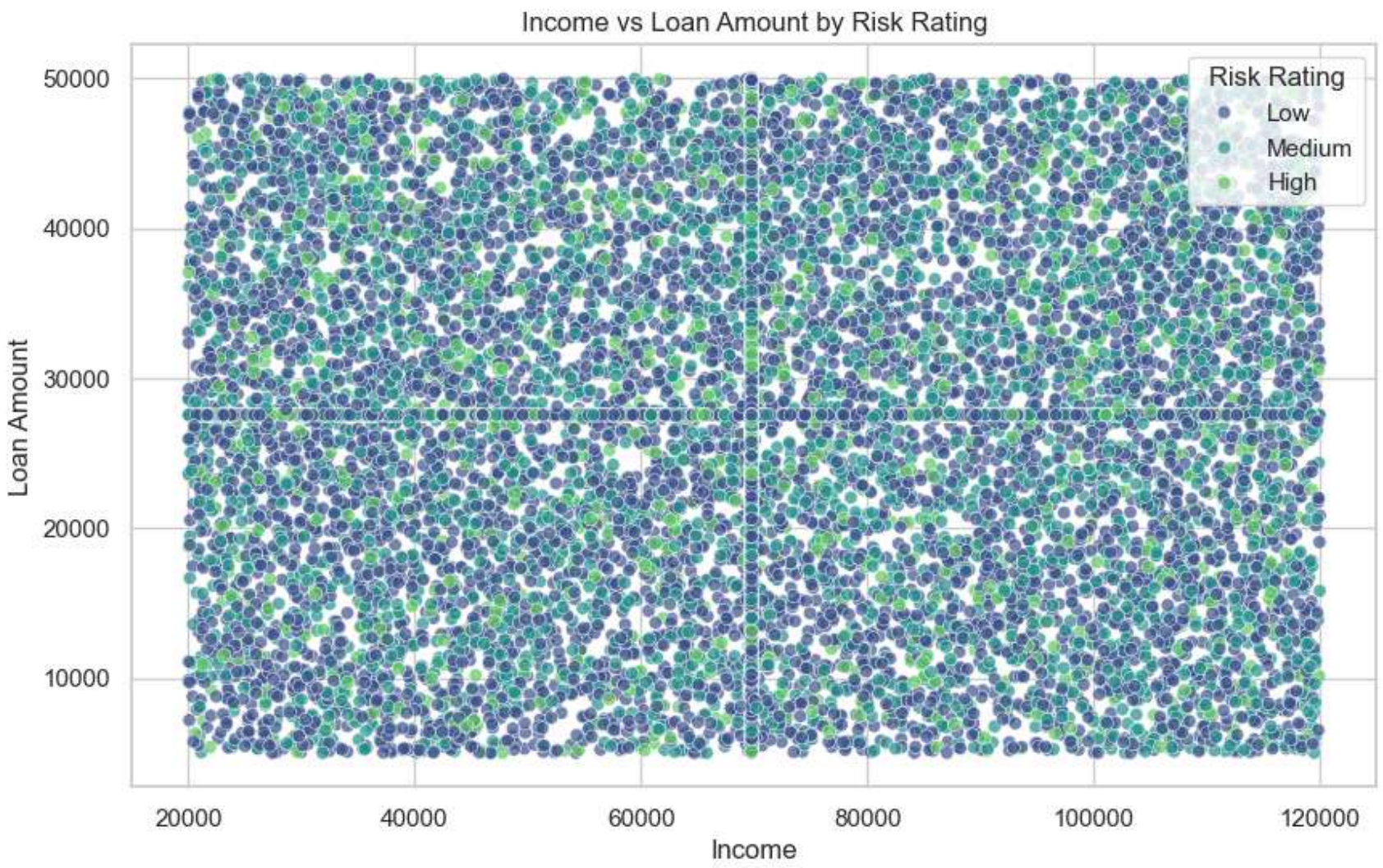
```
C:\Users\DEBABRATA\AppData\Local\Temp\ipykernel_16068\3730079387.py:9: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `h
ue` and set `legend=False` for the same effect.

  sns.boxplot(x='Risk Rating', y='Credit Score', data=df, palette='Set2')
```


Credit Score Distribution across Risk Ratings
Credit Score
800
775
750
725
700
675
650
625
600
Low
Medium
High
Risk Rating


Correlation Heatmap of Financial Features
Income
Credit Score
Loan Amount
Assets Value
Number of Dependents
Previous Defaults
1.00 0.01 -0.01 -0.01 0.01 -0.00
0.01 1.00 -0.01 0.01 0.01 -0.01
-0.01 -0.01 1.00 -0.01 0.00 0.01
-0.01 0.01 -0.01 1.00 -0.00 0.00
0.01 0.01 0.00 -0.00 1.00 -0.01
-0.00 -0.01 0.01 0.00 -0.01 1.00
Income
Credit Score
Loan Amount
Assets Value
er of Dependents
Previous Defaults
1.0
0.8
0.6
0.4
0.2
0.0



In [12]:
```
#Phase 5: Predictive Modeling (Machine Learning for Risk Assessment)
```

In [13]:
```
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report, accuracy_score
from sklearn.preprocessing import LabelEncoder
```

```
# Prepare features (X) and target (y)
# Let's encode the target variable 'Risk Rating'
le = LabelEncoder()
y = le.fit_transform(df['Risk Rating'])

# Select features (dropping target and non-predictive/high-cardinality text columns if needed)
X = df.drop(columns=['Risk Rating', 'City', 'State', 'Country'])
X = pd.get_dummies(X, drop_first=True)

# Train-Test Split (80% train, 20% test)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Initialize and train Random Forest Classifier
rf_model = RandomForestClassifier(n_estimators=100, random_state=42)
rf_model.fit(X_train, y_train)

# Predict and evaluate
y_pred = rf_model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred, target_names=le.classes_))

# Feature Importance
feature_importances = pd.Series(rf_model.feature_importances_, index=X.columns).nlargest(10)
plt.figure(figsize=(10, 6))
feature_importances.plot(kind='barh', color='teal')
plt.title('Top 10 Feature Importances for Risk Prediction')
plt.xlabel('Importance Score')
plt.savefig('feature_importances.png')
plt.show()
```

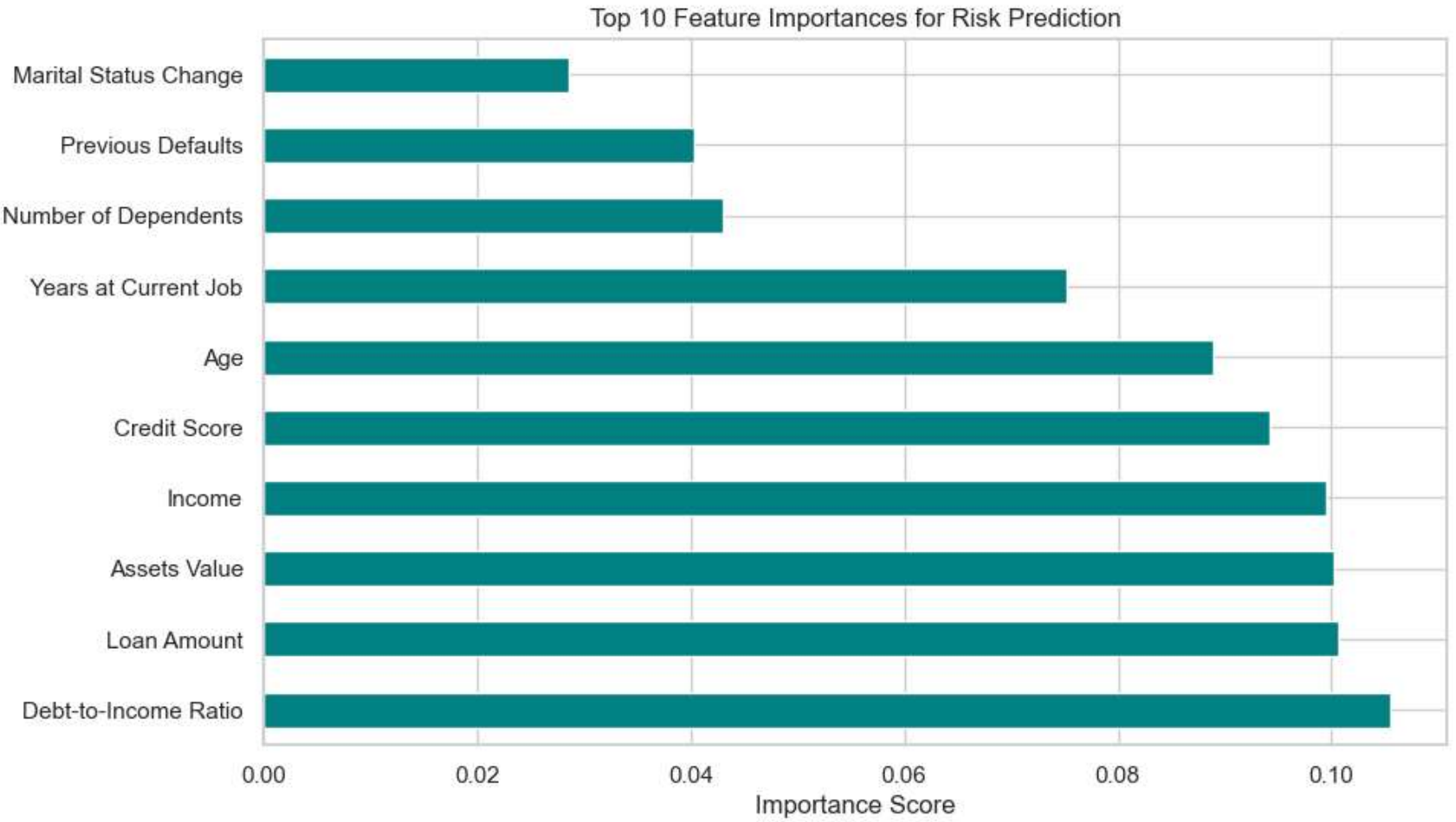
```
Accuracy: 0.594

Classification Report:
              precision    recall  f1-score   support

        High       0.00      0.00      0.00       326
         Low       0.60      0.99      0.74      1779
      Medium       0.47      0.03      0.05       895

    accuracy                           0.59      3000
   macro avg       0.36      0.34      0.26      3000
weighted avg       0.49      0.59      0.46      3000
```

```
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\metrics\_classification.py:1731: UndefinedMetricWarning: Precision
is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control thi
s behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\metrics\_classification.py:1731: UndefinedMetricWarning: Precision
is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control thi
s behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\metrics\_classification.py:1731: UndefinedMetricWarning: Precision
is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero_division` parameter to control thi
s behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
```



In [18]:
```
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.ensemble import RandomForestClassifier
from sklearn.preprocessing import LabelEncoder

# 2. Prepare Target and Features
le = LabelEncoder()
y = le.fit_transform(df['Risk Rating']) # Encodes Risk Rating into numeric classes

# Drop target and high-cardinality geographic columns for this model
X = df.drop(columns=['Risk Rating', 'City', 'State', 'Country'])
```

```
# One-hot encode remaining categorical variables
X = pd.get_dummies(X, drop_first=True)

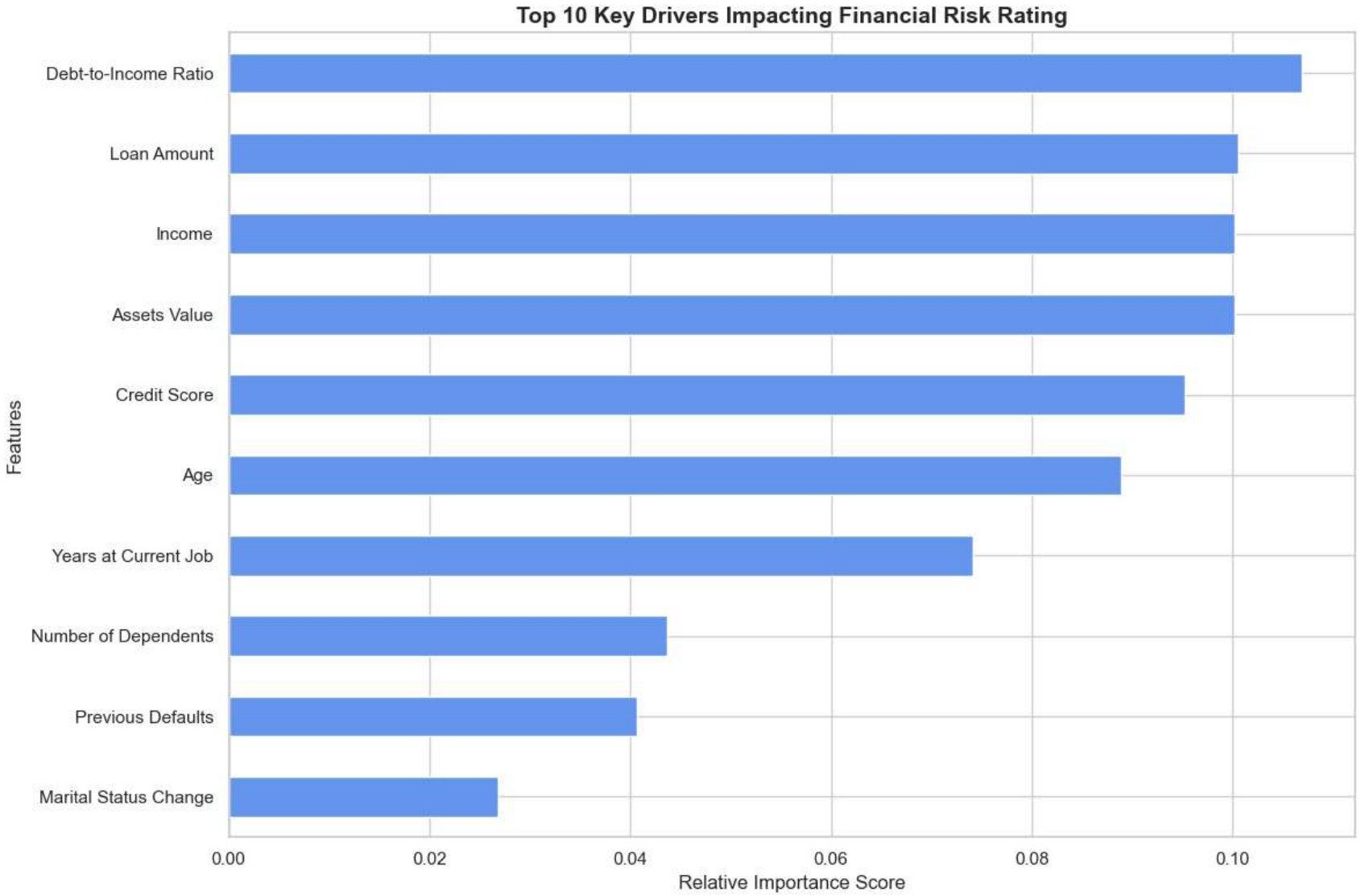
# 3. Train Random Forest to capture non-linear feature relationships
rf_model = RandomForestClassifier(n_estimators=150, random_state=42)
rf_model.get_params()
rf_model.fit(X, y)

# 4. Extract and Rank Feature Importances
feature_importances = pd.Series(rf_model.feature_importances_, index=X.columns).sort_values(ascending=False)

print("--- Top 15 Key Drivers of Financial Risk ---")
print(feature_importances.head(15))

# 5. Visualize Key Drivers
plt.figure(figsize=(12, 8))
feature_importances.head(10).plot(kind='barh', color='cornflowerblue')
plt.title('Top 10 Key Drivers Impacting Financial Risk Rating', fontsize=14, fontweight='bold')
plt.xlabel('Relative Importance Score', fontsize=12)
plt.ylabel('Features', fontsize=12)
plt.gca().invert_yaxis() # Highest importance at the top
plt.tight_layout()
plt.savefig('key_risk_drivers.png', dpi=300)
plt.show()
```

```
--- Top 15 Key Drivers of Financial Risk ---
Debt-to-Income Ratio               0.106856
Loan Amount                        0.100547
Income                             0.100225
Assets Value                       0.100149
Credit Score                       0.095246
Age                                0.088835
Years at Current Job               0.074113
Number of Dependents               0.043632
Previous Defaults                  0.040619
Marital Status Change              0.026865
Employment Status_Unemployed       0.015419
Employment Status_Self-employed    0.015290
Education Level_High School        0.014172
Loan Purpose_Home                  0.014122
Education Level_PhD                0.014120
dtype: float64
```


Top 10 Key Drivers Impacting Financial Risk Rating
Debt-to-Income Ratio
Loan Amount
Income
Assets Value
Credit Score
Age
Years at Current Job
Number of Dependents
Previous Defaults
Marital Status Change
Features
0.00
0.02
0.04
0.06
0.08
0.10
Relative Importance Score

In [ ]: